

Assessed Problem Sheet 1 - Week 2 (2025)

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This problem sheet is out of 36 marks. Your mark will be rescaled to a mark out of 100 when uploaded to Moodle.

1. Consider a quantum system with two orthogonal and normalised energy eigenstates $|\psi_1\rangle$ and $|\psi_2\rangle$. In answering this question, you may use Dirac notation or an appropriate column vector representation.

- (a) In Dirac notation, a ket vector $|\psi\rangle$ is normalised when $\langle\psi|\psi\rangle = 1$. Consider the state $|\alpha\rangle$, defined as

$$|\alpha\rangle = A (2|\psi_1\rangle - 5|\psi_2\rangle),$$

where A is real and non-negative. Compute the value of A that ensures that $|\alpha\rangle$ is normalised.

[4]

Solution: There are two possible methods, using Dirac notation or a column vector notation, and both are acceptable.

Dirac notation:

$$\begin{aligned}\langle\alpha|\alpha\rangle &= A^* (2\langle\psi_1| - 5\langle\psi_2|) A (2|\psi_1\rangle - 5|\psi_2\rangle) \\ &= |A|^2 (4\langle\psi_1|\psi_1\rangle - 10\langle\psi_1|\psi_2\rangle - 10\langle\psi_2|\psi_1\rangle + 25\langle\psi_2|\psi_2\rangle).\end{aligned}$$

Using orthonormality, $\langle\psi_i|\psi_j\rangle = \delta_{ij}$:

$$\langle\alpha|\alpha\rangle = |A|^2(4 + 25) = 29|A|^2.$$

Imposing normalisation $\langle\alpha|\alpha\rangle = 1$ gives $29|A|^2 = 1$, hence (since $A \geq 0$)

$$A = \frac{1}{\sqrt{29}}.$$

Column vectors:

$$A^* \begin{bmatrix} 2 & -5 \end{bmatrix} A \begin{bmatrix} 2 \\ -5 \end{bmatrix} = |A|^2(4 + 25) = 29|A|^2 \Rightarrow A = \frac{1}{\sqrt{29}}.$$

Marks: **1 mark** for correctly representing the inner product (either in Dirac notation or column vector). **2 marks** for the calculation. **1 mark** for a correct final answer.

- (b) Now consider the state $|\beta\rangle$, defined as

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$$|\beta\rangle = \sqrt{\frac{3}{4}}|\psi_1\rangle + \frac{e^{i\pi/6}}{2}|\psi_2\rangle.$$

Compute the inner product $\langle\beta|\alpha\rangle$, using the value of A computed above.

Hint: Remember to complex conjugate the coefficients when turning a ket into a bra.

Solution: We can work efficiently in column vectors (Dirac notation is also fine). First write the bra correctly complex conjugated:

$$\langle\beta| = [\sqrt{3}/2 \quad e^{-i\pi/6}/2].$$

Then

$$\begin{aligned}\langle\beta|\alpha\rangle &= [\sqrt{3}/2 \quad e^{-i\pi/6}/2] \cdot A \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \frac{1}{\sqrt{29}} \left(2 \cdot \frac{\sqrt{3}}{2} + (-5) \cdot \frac{e^{-i\pi/6}}{2} \right) \\ &= \frac{1}{\sqrt{29}} \left(\sqrt{3} - \frac{5}{2} e^{-i\pi/6} \right).\end{aligned}$$

Decimal (not required): $\approx -0.0804 + 0.2321 i$.

Marks: **1 mark** for writing down the inner product in either Dirac notation or column vector notation. **1 mark** for correctly complex conjugating the bra vector element. **1 mark** for the correct final answer (surd form is sufficient).

(c) The ket vector $|\gamma\rangle$ is of the form:

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$$|\gamma\rangle = x|\psi_1\rangle + y|\psi_2\rangle,$$

where x and y are real. You are told that $|\gamma\rangle$ is normalised and orthogonal to $|\alpha\rangle$. Use this information to calculate x and y . As a reminder, two states $|\psi\rangle$ and $|\phi\rangle$ are orthogonal when $\langle\psi|\phi\rangle = 0$.

Hint: Use orthogonality first to get a linear relation between x and y , then impose normalisation to fix their magnitudes.

Solution: We use the two pieces of information.

Orthogonality to $|\alpha\rangle$:

$$\langle\gamma|\alpha\rangle = 0 \quad \Rightarrow \quad (x\langle\psi_1| + y\langle\psi_2|) A (2|\psi_1\rangle - 5|\psi_2\rangle) = A(2x - 5y) = 0.$$

(1 mark)

Hence $2x = 5y \Rightarrow y = \frac{2}{5}x$. **(1 mark)**

Normalisation:

$$\langle\gamma|\gamma\rangle = x^2\langle\psi_1|\psi_1\rangle + y^2\langle\psi_2|\psi_2\rangle = x^2 + y^2 = 1.$$

Substitute $y = \frac{2}{5}x$:

$$x^2 \left(1 + \frac{4}{25} \right) = x^2 \frac{29}{25} = 1 \quad \Rightarrow \quad x = \pm \frac{5}{\sqrt{29}}. \text{ (1 mark)}$$

Then $y = \pm \frac{2}{\sqrt{29}}$ with the same overall sign. **(1 mark)**

Thus (either sign acceptable)

$$|\gamma\rangle = \frac{5}{\sqrt{29}}|\psi_1\rangle + \frac{2}{\sqrt{29}}|\psi_2\rangle. \textbf{(1 mark)}$$

Markers: Dirac or column vectors both acceptable.

- (d) Since $|\alpha\rangle$ and $|\gamma\rangle$ are orthogonal and normalised they represent an alternative orthonormal basis which can be used instead of $|\psi_1\rangle$ and $|\psi_2\rangle$. Therefore we can express $|\beta\rangle$ as:

$$|\beta\rangle = p|\alpha\rangle + q|\gamma\rangle,$$

where p and q are complex numbers. Calculate the values of p and q .

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Solution: There are two obvious methods, either using a basis change matrix or directly computing the coefficients.

Option 1: Direct computation

Because $\{|\alpha\rangle, |\gamma\rangle\}$ is orthonormal,

$$p = \langle\alpha|\beta\rangle, \quad q = \langle\gamma|\beta\rangle.$$

We already computed $\langle\beta|\alpha\rangle$ above, so

$$p = \langle\alpha|\beta\rangle = (\langle\beta|\alpha\rangle)^* = \frac{1}{\sqrt{29}} \left(\sqrt{3} - \frac{5}{2}e^{+i\pi/6} \right) \approx -0.0804 - 0.2321i.$$

For q , use the $|\psi_1\rangle, |\psi_2\rangle$ representation of $|\gamma\rangle$:

$$\langle\gamma| = \left[\frac{5}{\sqrt{29}} \quad \frac{2}{\sqrt{29}} \right], \quad |\beta\rangle = \begin{bmatrix} \sqrt{3}/2 \\ e^{i\pi/6}/2 \end{bmatrix}.$$

Hence

$$q = \langle\gamma|\beta\rangle = \frac{1}{\sqrt{29}} \left(\frac{5\sqrt{3}}{2} + \frac{e^{-i\pi/6}}{2} \right) \approx 0.9649 - 0.0928i.$$

Option 2: Use a change of basis matrix

Students could also use the similarity matrix:

$$S = \begin{bmatrix} \langle\alpha|\psi_1\rangle & \langle\alpha|\psi_2\rangle \\ \langle\gamma|\psi_1\rangle & \langle\gamma|\psi_2\rangle \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{29}} & \frac{-5}{\sqrt{29}} \\ \frac{5}{\sqrt{29}} & \frac{2}{\sqrt{29}} \end{bmatrix}, \quad \begin{bmatrix} p \\ q \end{bmatrix} = S \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \text{ with } \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 \\ e^{i\pi/6}/2 \end{bmatrix}.$$

This reproduces the same p, q as above.

Marks: Because there are multiple ways to solve this, the mark scheme is not so finely defined. **4 marks** for correct numerical values of p and q and a clear and correct calculation. If there are minor numerical slips, award **3 marks**. If there are only part of the correct method award **1-2** marks according to how far towards the correct method the student has got and whether there are also calculation errors. It is **not** necessary to convert a surd form solution to a decimal to gain full marks, however students should have simplified their answer as much as possible.

2. Spin is a property of quantum particles which you will learn about in chapters 3 and 4 of this course.

The state of a spin-half system is described by a two-dimensional vector space with the orthonormal basis $|\uparrow\rangle$ and $|\downarrow\rangle$ which we will represent using the column-vector representation:

$$|\uparrow\rangle \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |\downarrow\rangle \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A nuclear spin-half system is subject to a magnetic field. This is characterised by a Hamiltonian $\hat{H}$ with the following matrix representation H with respect to the $|\uparrow\rangle, |\downarrow\rangle$ basis:

$$H = E \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = E \sigma_y, \quad (1)$$

where E has units of energy and depends on the strength of the field.

The normalised eigenvectors of H have the form:

$$|+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + e^{i\epsilon}|\downarrow\rangle) \quad |-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - e^{i\epsilon}|\downarrow\rangle), \quad (2)$$

where ϵ is a real constant you will determine.

- (a) Compute the eigenvalues of H and then the eigenvectors (expressing these in the form of equation (2) but calculating the value of ϵ). Clearly state which eigenvector is associated with which eigenvalue. [4]

Solution: We compute the eigenvalues via the characteristic equation:

$$\det \begin{bmatrix} -\lambda & -iE \\ iE & -\lambda \end{bmatrix} = (-\lambda)(-\lambda) - (-iE)(iE) = \lambda^2 - E^2 = 0.$$

Hence $\lambda = \pm E$. **(2 marks)**

Now solve the eigenvalue equation $H|\pm\rangle = \pm E|\pm\rangle$ using the Ansatz in (2). Write

$$|\pm\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \pm e^{i\epsilon} \end{bmatrix}.$$

Act with H :

$$H|\pm\rangle = E \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \pm e^{i\epsilon} \end{bmatrix} = \frac{E}{\sqrt{2}} \begin{bmatrix} \mp i e^{i\epsilon} \\ i \end{bmatrix}.$$

Impose $H|+\rangle = +E|+\rangle$:

$$\frac{E}{\sqrt{2}} \begin{bmatrix} -i e^{i\epsilon} \\ i \end{bmatrix} = \frac{E}{\sqrt{2}} \begin{bmatrix} 1 \\ e^{i\epsilon} \end{bmatrix} \Rightarrow \begin{cases} -i e^{i\epsilon} = 1, \\ i = e^{i\epsilon}. \end{cases}$$

Both give $e^{i\epsilon} = i \Rightarrow \epsilon = \pi/2 \pmod{2\pi}$. Therefore

$$|+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle) \quad (\text{eigenvalue } +E), \quad |-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - i|\downarrow\rangle) \quad (\text{eigenvalue } -E).$$

(2 marks) Marks: 1 mark for each eigenvalue (**2 marks** in total) and 1 mark for associating each correctly with $|+\rangle$ and $|-\rangle$ (**2 marks** in total).

- (b) You are given a state, $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ and need to express $|\psi\rangle$ in the $|\pm\rangle$ basis, i.e. $|\psi\rangle = \alpha'|+\rangle + \beta'|-\rangle$. Using the expression given in the lecture notes for its matrix elements, derive the similarity transform matrix S which relates α, β and α', β' :

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$$\begin{bmatrix} \alpha' \\ \beta' \end{bmatrix} = S \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Verify that your S matrix is unitary.

Solution: From the general form,

$$\begin{bmatrix} \alpha' \\ \beta' \end{bmatrix} = \begin{bmatrix} \langle + | \uparrow \rangle & \langle + | \downarrow \rangle \\ \langle - | \uparrow \rangle & \langle - | \downarrow \rangle \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Compute the four inner products using $|+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - i|\downarrow\rangle)$:

$$\begin{aligned} \langle + | \uparrow \rangle &= \frac{1}{\sqrt{2}}, & \langle + | \downarrow \rangle &= \frac{(-i)^*}{\sqrt{2}} = \frac{-i}{\sqrt{2}}, \\ \langle - | \uparrow \rangle &= \frac{1}{\sqrt{2}}, & \langle - | \downarrow \rangle &= \frac{(i)^*}{\sqrt{2}} = \frac{i}{\sqrt{2}}. \end{aligned}$$

Hence

$$S = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix}.$$

(4 marks for the four inner products; 1 mark for S .)

Check unitarity explicitly:

$$\begin{aligned} S^\dagger &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} \\ SS^\dagger &= \frac{1}{2} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 + (-i)i & 1 + (-i)(-i) \\ 1 + i \cdot i & 1 + i(-i) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Thus S is unitary. **(1 mark)**

- (c) Use the similarity matrix you have derived to transform the H matrix above into a matrix representation in the $|+\rangle, |-\rangle$ basis.

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Solution: Let H' be the representation of $\hat{H}$ in the $\{|+\rangle, |-\rangle\}$ basis. From the notes:

$$H' = SHS^\dagger.$$

Substitute the matrices:

$$H' = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} E \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}.$$

Carrying out the multiplication (or, equivalently, using $H|\pm\rangle = \pm E|\pm\rangle$ to read off matrix elements $\langle m|H|n\rangle$) yields

$$H' = E \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Marks: 1 mark for the correct equation to calculate H' , **2 marks** for a correct matrix product (including correctly computing $S^\dagger$) and **1 mark** for the correct H' matrix.

3. You can apply the same techniques you used in previous QM courses to solve the Time-Dependent Schrödinger Equation (TDSE) for wave-functions to study time evolution in Dirac notation.

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When the spin system from Q2 evolves in time, its states evolve under the TDSE:

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = \hat{H} |\psi(t)\rangle.$$

The following state (expressed in the $|+\rangle$, $|-\rangle$ basis) is a general solution of the TDSE with Hamiltonian $\hat{H}$ from equation (1) where ϕ and θ are constants that you will identify,

$$|\psi(t)\rangle = ae^{i\phi t}|+\rangle + be^{i\theta t}|-\rangle.$$

By substituting $|\psi(t)\rangle$ into the TDSE determine the values of ϕ and θ .

Solution: Left-hand side (LHS): Differentiate term by term:

$$\frac{\partial}{\partial t} |\psi(t)\rangle = a(i\phi)e^{i\phi t}|+\rangle + b(i\theta)e^{i\theta t}|-\rangle.$$

Multiply by $i\hbar$:

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = i\hbar [a(i\phi)e^{i\phi t}|+\rangle + b(i\theta)e^{i\theta t}|-\rangle] = -\hbar [a\phi e^{i\phi t}|+\rangle + b\theta e^{i\theta t}|-\rangle].$$

Right-hand side (RHS): Using $H|\pm\rangle = \pm E|\pm\rangle$ from Q2,

$$\hat{H} |\psi(t)\rangle = ae^{i\phi t}(+E)|+\rangle + be^{i\theta t}(-E)|-\rangle = Eae^{i\phi t}|+\rangle - Ebe^{i\theta t}|-\rangle.$$

Equate coefficients of the independent basis vectors $|+\rangle$ and $|-\rangle$:

$$-\hbar\phi = E, \quad -\hbar\theta = -E.$$

Therefore

$$\phi = -\frac{E}{\hbar}, \quad \theta = \frac{E}{\hbar}.$$

Marks: 2 marks for the correct expression on the LHS, **2 marks** for the correct RHS, **2 marks** for the correct values of ϕ and θ .